Novel Feature Selection Method for Single Cell RNA Sequencing Data

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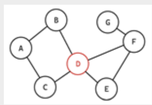
Abstract

Feature selection plays an important role in high dimensional data analyzes, allowing for scalability and noise reduction. For single cell RNA-Seq (scRNA-seq) data, feature/gene selection helps scale these high dimensional datasets for complex downstream analyses as well as distinguish 'informative' genes typically marking cell types or states and identifying unique populations. To date, only a few upstream feature selection methods exist. The most common method to exclude noisy or uninformative genes and features from scRNA-Seq is by prioritizing the most variable genes based on mean and variance or dispersion of expression levels, followed by gene selection using high average TF-IDF score genes [3]. Several probabilistic approaches such as differential gene expression analysis or deep learning based feature importance prioritization identify marker genes for identified cell populations. These methods, however, require existing knowledge or assumptions regarding clusters or labels and are applied *after* cell type identification or clustering. Here, we propose a novel algorithm for feature selection by utilizing a robust graph-based approach. Furthermore, our approach provides biological interpretability of the gene selection process allowing for enhanced and scalable downstream analyzes, especially for clustering methods. To validate our results, we contrast the two existing methods, in particular highly variable genes and TF-IDF based selection, to our novel graph based approach. We test multiple real publicly available scRNA-Seq data for accuracy in downstream results such as clustering, cell type, state identification, and trajectory analysis among other analyses. Our preliminary results suggest our novel approach introduces a strength to the standard TF-IDF based approach and outperforms feature selection using highly variable genes by multiple metrics by identifying more biologically informative features while reducing the overall feature noise.

Background

We propose a graph based method that focuses on the concept of betweenness centrality [2], a score that quantifies the importance of a node in a network depending on the extent it lies in the path between other nodes.

$$C_B(v) = \sum_{s \neq v \neq t \in V} \frac{\sigma_{st}(v)}{\sigma_{st}}$$



Following this intuition, when graphing genes as nodes, the edge between each gene holds a score/weight which determines how strong the similarity or relationship is between the two connected genes. Therefore genes with many 'strong' connections have the highest betweenness centrality and are considered the most significant/informative gene.

Method

For our experimental test of selecting **Central Genes** as highly informative features, we used a collection of 3 datasets.

1. Zheng et al. (2017), a 10x Genomics PBMCs dataset, composed of 7174 genes and 2882 cells. This dataset is already labeled and is used as our "ground truth".
2. Gubin et al. (2018), a 10x Genomics aCTLA4 dataset, composed of 11501 genes and 3148 cells.
3. Liao et al. (2020), a 10x Genomics COVID-19 dataset, composed of 12045 genes and 7158 cells.

For each dataset, we filtered genes based on Gene Symbol convention (removing genes that follow conventions such as IGHV/IGKV, TRAV/TRBV, RPL/RPS, MT-/GM, RIK). We apply additional preprocessing for speed optimization which includes filtering out genes with low variability to subset the genes for each dataset. Next, we calculate cosine distance adjacency matrix and impose a dynamically set (data-learned) cutoff for each subset. A graph object is created with the adjacency matrix followed by finding the betweenness centrality metric of the graph (our implementation uses igraph [6] R package). The node with the highest betweenness score is stored then removed from the graph. This process is repeated n times, calculating the betweenness and removing/registering the highest scoring node/gene constructing a set of n selected Central Genes.
To assess our results we applied the following assessment metrics:

- Overlap with highly variable genes and TF-IDF.
- Overlap with downstream DE genes analysis (with PBMCs)
- Clustering Performance in Downstream (with PBMCs)

Results

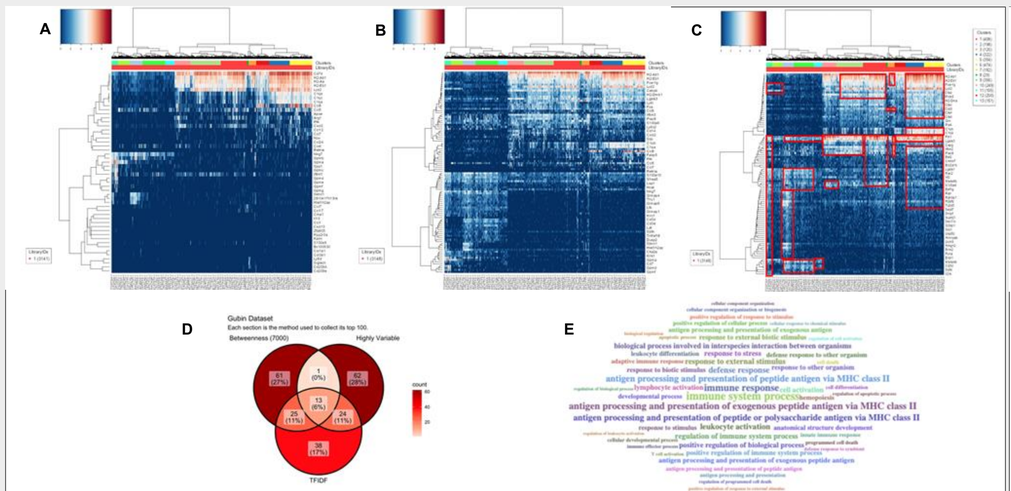


Figure 1: Methods being applied to Gubin dataset. **A**) Heatmap of Highly Variable Genes (HVG) selected top 100 from Gubin dataset. **B**) Heatmap of TF-IDF selected top 100 from Gubin dataset. **C**) Heatmap of Central Genes selected top 100 from Gubin dataset. **D**) Venn diagram comparing the genes selected by HVG, TF-IDF, and Central Genes. **E**) Gene enrichment of genes from Central Genes.

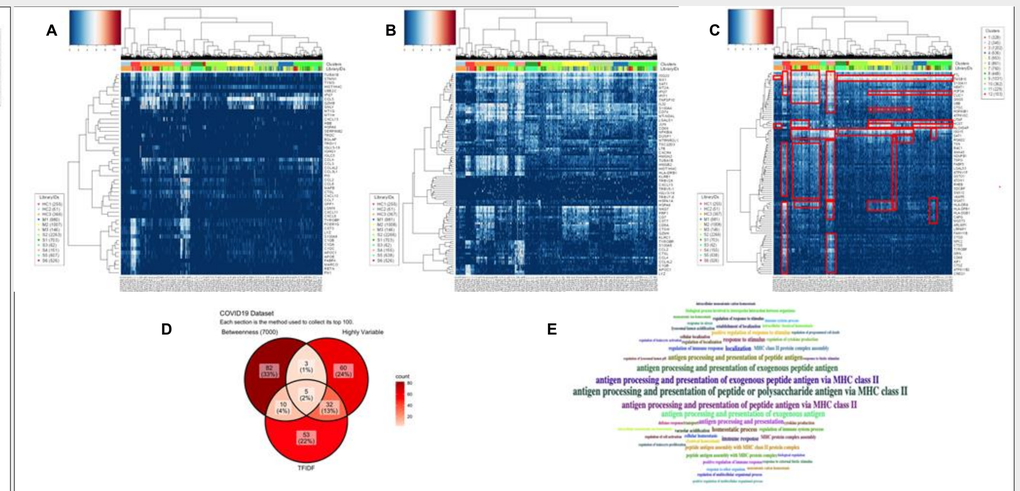


Figure 2: Methods being applied to COVID19 dataset. **A**) Heatmap of Highly Variable Genes (HVG) selected top 100 genes from COVID19 dataset. **B**) Heatmap of TF-IDF selected top 100 genes from COVID19 dataset. **C**) Heatmap of Central Genes selected top 100 genes from COVID19 dataset. **D**) Venn diagram comparing the genes selected by HVG, TF-IDF, and Central Genes. **E**) Gene enrichment of genes from Central Genes.

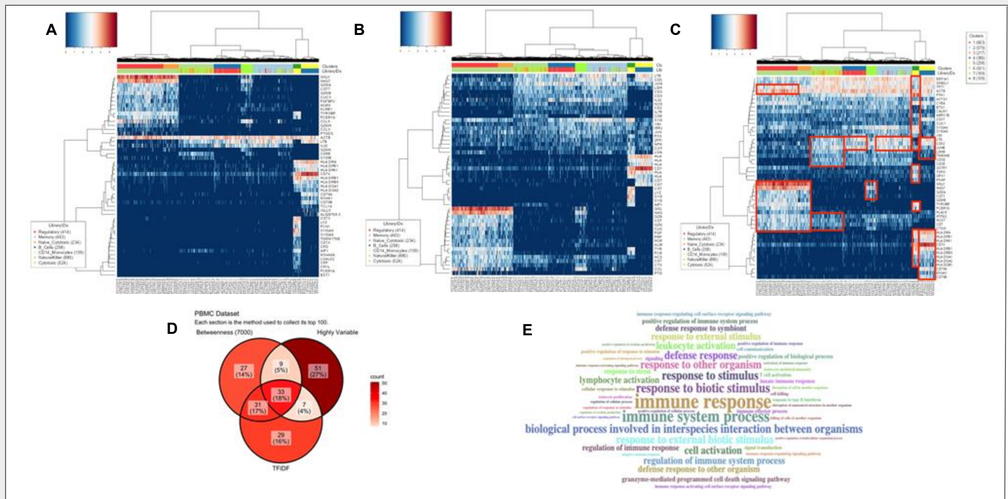


Figure 3: Methods being applied to PBMC dataset. **A**) Heatmap of Highly Variable Genes (HVG) selected top 50 genes from PBMC dataset. **B**) Heatmap of TF-IDF selected top 50 genes from PBMC dataset. **C**) Heatmap of Central Genes selected top 50 genes from PBMC dataset. **D**) Venn diagram comparing the genes selected by HVG, TF-IDF, and Central Genes. **E**) Gene enrichment of genes from Central Genes.

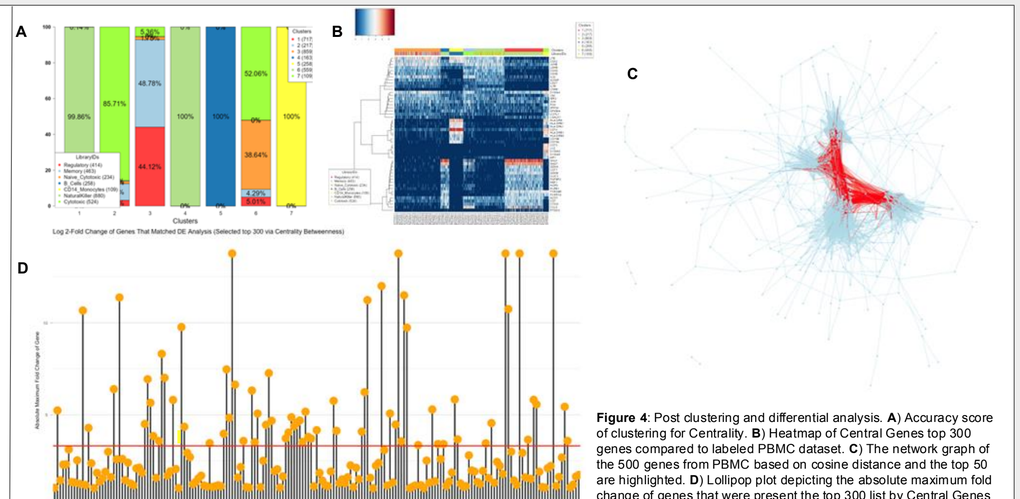


Figure 4: Post clustering and differential analysis. **A**) Accuracy score of clustering for Centrality. **B**) Heatmap of Central Genes top 300 genes compared to labeled PBMC dataset. **C**) The network graph of the 500 genes from PBMC based on cosine distance and the top 50 are highlighted. **D**) Lollipop plot depicting the absolute maximum fold change of genes that were present the top 300 list by Central Genes compared to the PBMC DE Analysis from SC1 [1].

Discussion

Our analysis indicates that Central Genes is a comparable or in some cases a better pre-clustering method than highly variable genes and TF-IDF. For performance we are applying a layer of preprocessing that selects the top 7000 highly variable genes for our Central Genes method. From the observations that can be made about **Figures 1c, 2c, 3c**, when only considering the top 7000 we are performing significantly better in terms of identifying gene modules compared to their highly variable gene counterparts. When comparing to TF-IDF counterparts, they appear to have a large number of similarities which is supported by **Figures 1d, 2d, 3d** which is why their heat maps have some similarities however in some cases between **Figure 1b & 1c** and **Figure 3b & 3c** Central Genes appears to be able to display more separated gene modules. Moreover, when we analyze **Figure 4a**, we see that Central Genes has a high accuracy score in clustering over the labeled PBMC dataset which can be further seen in **Figure 4b**. To determine if the number genes being selected are informative, our DE analysis with SC1 [1] for the PBMC dataset had 476/7174 genes considered as informative. When considering all three methods' ability to select the top 300, highly variable genes matched 151 genes, TF-IDF matched 145 genes, and Central Genes matched 187 genes as depicted in **Figure 4d**.

Conclusion

In conclusion, our study shows Betweenness Centrality as a comparable technique to current pre-clustering methods while making minimal model assumptions. In most cases, it outperformed highly variable genes and in some cases it outperformed TF-IDF. These findings suggest that this novel method can be a standalone feature selection as well as a combination with other methods to form a ensemble method which combines several methods together to build off each methods' advantages while reducing the disadvantages. Future research should explore scalability to larger gene counts and how it would affect performance on speed and accuracy.

References

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